# Midterm Exam II — Part 1

PPEM 454 — Viral Ecology

**Name:**

_____

**Date:**

_____

**Workflow:** Host selection, sequences, alignment, similarity, phylogenetics, and k-mers.

**Before knitting:** Install `ViralExamR`, `ape`, and `phangorn` in the R console, not in this document. Select the desired host by editing `params: host:` in the YAML header; valid values depend on `ViralExamR::get_sequences()`. The interactive `choose_host()` menu cannot run during knitting.

## 1. Load packages

```
library(ViralExamR)
library(ape)
library(phangorn)
```

## 2. Choose a host and load sequences

**Q0. Write your selected host.**

---

```
host <- "plant" # Set in YAML: plant, fungus, bacteria, animal, or human
sequences <- get_sequences(host)
print(data.frame(sequence = names(sequences), length = nchar(sequences)))
```

```
##
## KU687048 Banana bunchy top virus isolate Menk-22 segment DNA-R, complete sequence KU687048 Banana bunch
## KU687046 Banana bunchy top virus isolate Bit-20 segment DNA-R, complete sequence   KU687046 Banana bunc
## KU687040 Banana bunchy top virus isolate Lubum-4 segment DNA-R, complete sequence KU687040 Banana bunch
## KU687045 Banana bunchy top virus isolate Kwg-19 segment DNA-R, complete sequence   KU687045 Banana bunc
## KU687042 Banana bunchy top virus isolate Kip-15 segment DNA-R, complete sequence   KU687042 Banana bunc
## KU687041 Banana bunchy top virus isolate Lubum-9 segment DNA-R, complete sequence KU687041 Banana bunch
## KR350618 Banana bunchy top virus isolate Trp-DH8 segment DNA-R, complete sequence KR350618 Banana bunch
## AF416468 Banana bunchy top virus from Taiwan DNA-1 segment, complete sequence         AF416468 Banana b
##                                                                                                  length
## KU687048 Banana bunchy top virus isolate Menk-22 segment DNA-R, complete sequence   1127
## KU687046 Banana bunchy top virus isolate Bit-20 segment DNA-R, complete sequence    1127
## KU687040 Banana bunchy top virus isolate Lubum-4 segment DNA-R, complete sequence   1127
## KU687045 Banana bunchy top virus isolate Kwg-19 segment DNA-R, complete sequence    1127
## KU687042 Banana bunchy top virus isolate Kip-15 segment DNA-R, complete sequence    1127
## KU687041 Banana bunchy top virus isolate Lubum-9 segment DNA-R, complete sequence   1127
## KR350618 Banana bunchy top virus isolate Trp-DH8 segment DNA-R, complete sequence   1127
## AF416468 Banana bunchy top virus from Taiwan DNA-1 segment, complete sequence       1127
```

## 3. Align and inspect sequences

Choose an alignment method: `ClustalW`, `ClustalOmega`, or `Muscle`.

```
alignment <- align_sequences(sequences, method = params$alignment_method)
```

```
## use default substitution matrix
```

```
print(alignment)
```

```
##
## "---------------TATTACCCCCAGCGCTCGGGACGGGACATTTGCATCTATAAATAGACCTCCCCCCTCTCCATTACAAGATCATCATCGACGACAGAA
##
## "---------------TATTACCCCCAGCGCTCGGGACGGGACATTTGCATCTATAAATAGACCTCCCCCCTCTCCATTACAAGATCATCATCGACGACAGAA
##
## "---------------TATTACCCCCAGCGCTCGGGACGGGACATTTGCATCTATAAATAGACCTCCCCCCTCTCCATTACAAGATCATCATCGACGACAGAA
##
## "---------------TATTACCCCCAGCGCTCGGGACGGGACATTTGCATCTATAAATAGACCTCCCCCCTCTCCATTACAAGATCATCATCGACGACAGAA
##
## "---------------TATTACCCCCAGCGCTCGGGACGGGACATTTGCATTTATAAATAGACCTCCCCCCTCTCCATTACAAGATCATCATCGACGACAGAA
##
## "---------------TATTACCCCCAGCGCTCGGGACGGGACATTTGCATCTATAAATAGACCTCCCCCCTCTCCATTACAAGATCATCATCGACGACAGAA
##
## "---------------TATTACCCCCAGCGCTCGGGACGGGACATTTGCATCTATAAATAGACCTCCCCCCTCTCCAATACAAGATCATCATCGACGACAGAA
##
## "AGCGCTGGGGCTTATTATTACCCCCAGCGCTCGGGACGGGACATTTGCATCTATAAATAGACCTCCCCCCCCTCCACTACAAGATCATCATCGTCGACAGAA
```

## 4. Pairwise sequence identity

```
# Convert aligned sequences to ape's DNAbin format
alignment_matrix <- do.call(
  rbind,
  strsplit(chartr("U", "T", toupper(alignment)), "", fixed = TRUE)
)
dna <- ape::as.DNAbin(alignment_matrix)
# Raw distance is the proportion of differing bases.
raw_distances <- ape::dist.dna(
  dna,
  model = "raw",
  pairwise.deletion = TRUE
)
# Convert it to percentage identity; exclude gaps/ambiguities pairwise.
identity_matrix <- 100 * (1 - as.matrix(raw_distances))
diag(identity_matrix) <- 100

# Here are the pairwise sequence identity results
print(round(identity_matrix, 2))
```

```
##                                                                                 KU687048 Banana bunch
## KU687048 Banana bunchy top virus isolate Menk-22 segment DNA-R, complete sequence
## KU687046 Banana bunchy top virus isolate Bit-20 segment DNA-R, complete sequence
## KU687040 Banana bunchy top virus isolate Lubum-4 segment DNA-R, complete sequence
## KU687045 Banana bunchy top virus isolate Kwg-19 segment DNA-R, complete sequence
## KU687042 Banana bunchy top virus isolate Kip-15 segment DNA-R, complete sequence
## KU687041 Banana bunchy top virus isolate Lubum-9 segment DNA-R, complete sequence
## KR350618 Banana bunchy top virus isolate Trp-DH8 segment DNA-R, complete sequence
## AF416468 Banana bunchy top virus from Taiwan DNA-1 segment, complete sequence
##                                                                                 KU687046 Banana bunch
```

```
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```

The overall identity is the mean of pairwise identities. Gaps and ambiguous positions are excluded pairwise in the supplied calculation.

**Q1. What is the overall pairwise identity? Use `mean()`.**

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**Q2. Which pair is most and least similar, and what percentage identity do they share?**

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## 5. Neighbor-Joining tree

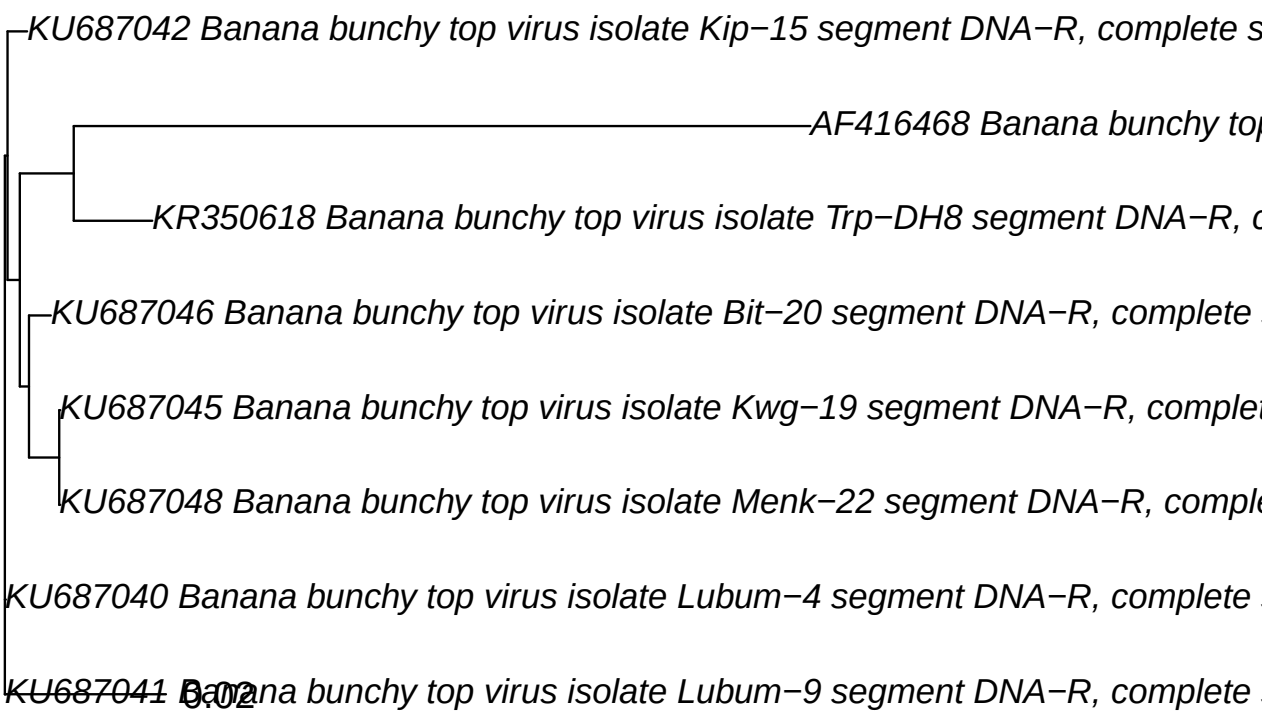
Select an evolutionary distance model, for example `RAW`, `JC69`, or `K80`.

```
distances <- ape::dist.dna(dna, model = params$nj_model, pairwise.deletion = FALSE)
# review the evolutionary distances
print(round(as.matrix(distances), 4))
```

```
##                                                                                 KU687048 Banana bunch
## KU687048 Banana bunchy top virus isolate Menk-22 segment DNA-R, complete sequence
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## KR350618 Banana bunchy top virus isolate Trp-DH8 segment DNA-R, complete sequence
## AF416468 Banana bunchy top virus from Taiwan DNA-1 segment, complete sequence
```

```r
# Calculate Neighbor-Joining tree
tree_nj <- ape::nj(distances)
# Plot Neighbor-Joining tree
plot(tree_nj, type = "phylogram", cex = 0.9,
     main = paste("Neighbor-Joining tree:", host, "virus"))
ape::add.scale.bar()
```

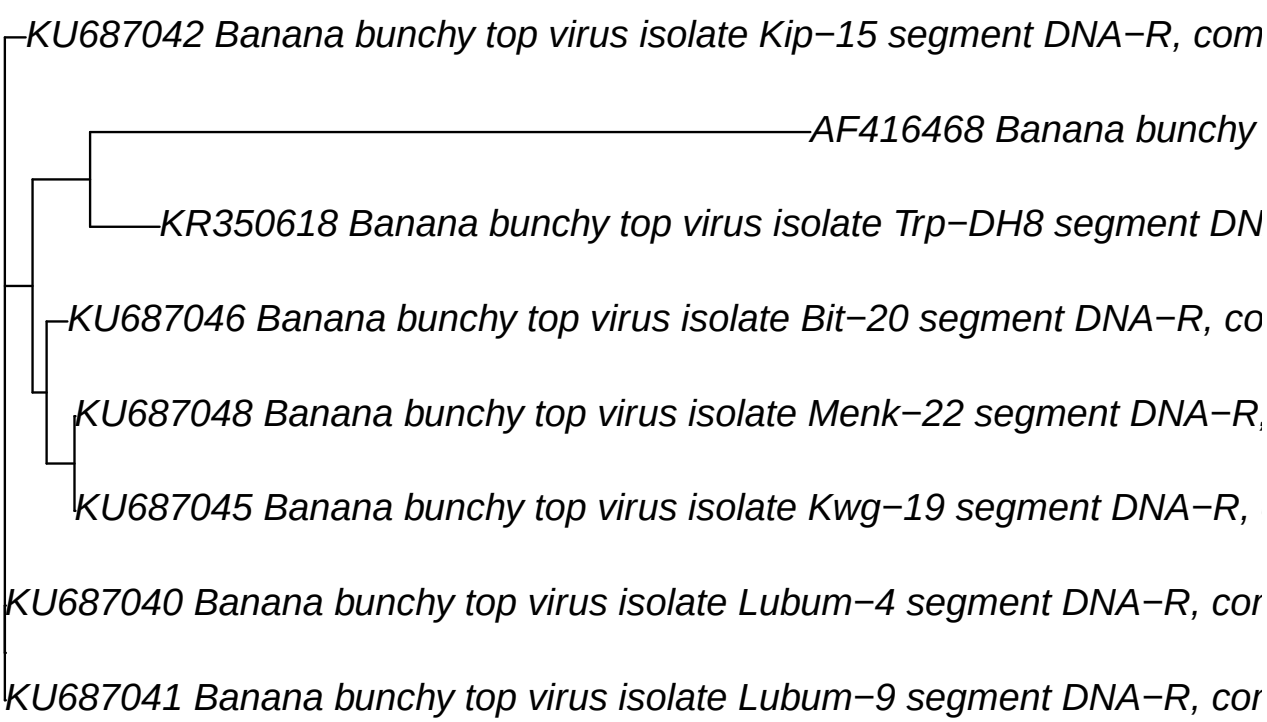



### 6. Maximum-Likelihood tree

```
phy_data <- phangorn::phyDat(dna, type = "DNA")
# Calculate Neighbor-Joining tree
fit <- phangorn::pml(tree_nj, data = phy_data)
fit_ml <- phangorn::optim.pml(fit, model = "GTR", optNni = TRUE,
                              optBf = FALSE, optQ = FALSE,
                              optGamma = FALSE, optInv = FALSE)
```

```
## optimize edge weights:  -2210.871 --> -2209.074
## optimize rate matrix:  -2209.074 --> -2195.508
## optimize edge weights:  -2195.508 --> -2195.505
## optimize topology:  -2195.505 --> -2195.505  NNI moves:  0
## optimize rate matrix:  -2195.505 --> -2195.505
## optimize edge weights:  -2195.505 --> -2195.505
```

```
tree_ml <- fit_ml$tree
# Plot Maximum-Likelihood tree
plot(tree_ml, type = "phylogram", main = paste("Maximum-Likelihood tree:", host, "virus"))
```




**Q4. Do the most similar sequences cluster together?**

**Q5. Are the tree topologies from NJ and ML identical? Why?**

**Q6. What happens if you change the evolutionary model in NJ and ML?**

**Q7. Do you notice a change in branch lengths? Why?**

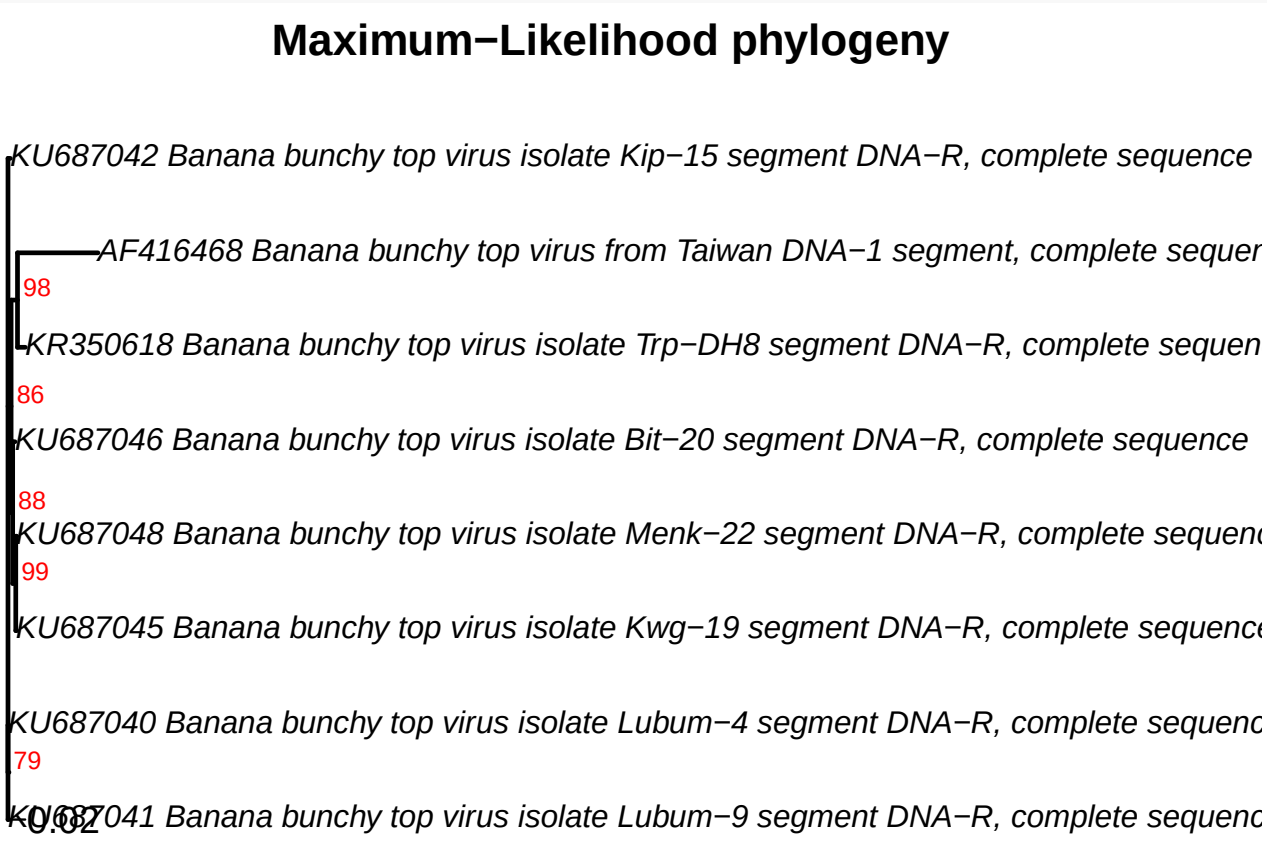
## 7. Bootstrapping

The following implements the original 100-replicate procedure. **Note:** It generates NJ bootstrap trees (K80) and maps their bipartition support onto the ML reference topology; it is not an ML bootstrap.

```
bsml <- ape::boot.phylo(
  phy = tree_ml,
  x = dna,
  FUN = function(x) {
    ape::nj(ape::dist.dna(x, model = "K80"))
  },
  B = 100,
  trees = TRUE
)
```

```
## Running bootstraps:       100 / 100
## Calculating bootstrap values... done.
```

```
# Plot ML tree with bootstrap support
plot(
  tree_ml,
  main = "Maximum-Likelihood phylogeny",
  edge.width = 2,
  cex = .8
)
# Add bootstrap support
ape::nodelabels(
  text = bsml$BP,
  node = (Ntip(tree_ml) + 1):(Ntip(tree_ml) + tree_ml$Nnode),
  frame = "none",
  cex = 0.7,
  adj = c(-0.2, -0.2),
  col = "red"
)
ape::add.scale.bar()
```



**Q8. What is bootstrap support, and what do the support values on your tree mean?**

---

---

---

**Q9. How would you interpret low bootstrap support?**

---

---

---

## 8–9. k-mers

Select the two most divergent sequences using the identity matrix and compare their k-mer profiles (`k = 3` by default). Edit the selected indices if needed.

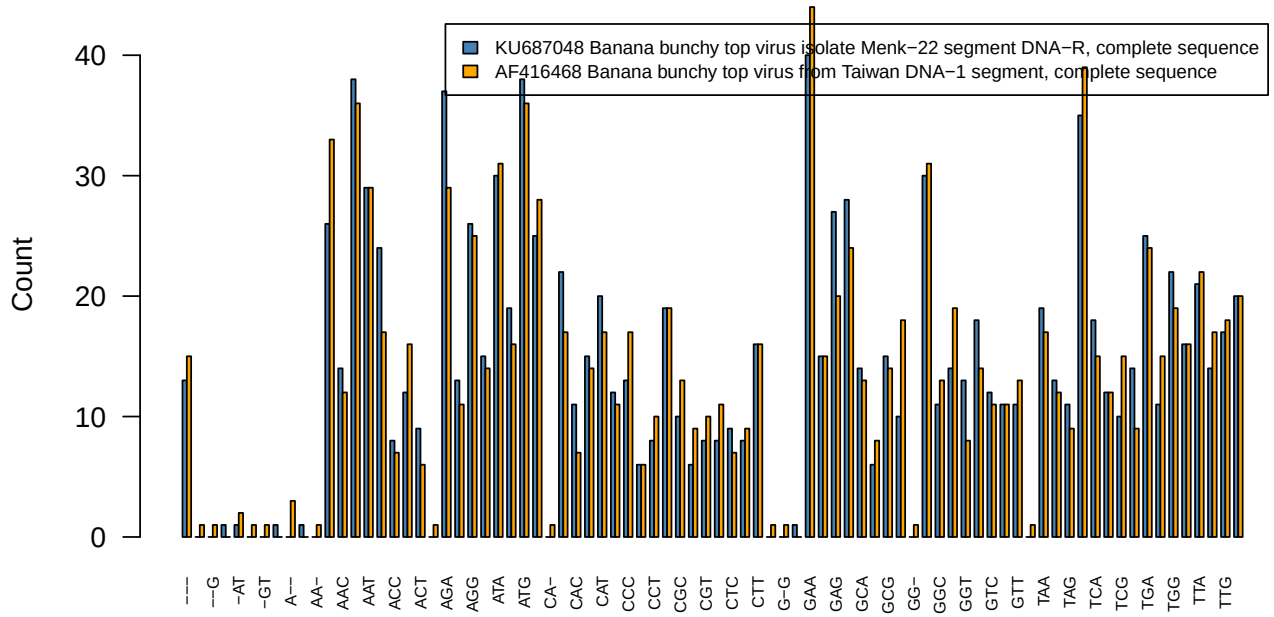
```
k <- params$k
# Choose the most divergent pair automatically from the pairwise matrix.
pairwise_id <- identity_matrix
pairwise_id[lower.tri(pairwise_id, diag = TRUE)] <- Inf
selected_pair <- arrayInd(which.min(pairwise_id), dim(pairwise_id))
selected_pair
```

```
##      [,1] [,2]
## [1,]    1    8
```

```
kmers_spp <- count_kmers(sequences[as.vector(selected_pair)], k)
a <- kmers_spp[[1]]
b <- kmers_spp[[2]]

# Align k-mers and replace missing counts with zero
kmers <- sort(union(names(a), names(b)))
counts <- rbind(a[kmers], b[kmers])
counts[is.na(counts)] <- 0

# Plotting k-mer sizing between a pair of sequences
barplot(
  counts,
  beside = TRUE,
  names.arg = kmers,
  col = c("steelblue", "orange"),
  las = 2,
  cex.names = 0.6,
  ylab = "Count",
  legend.text = c(names(kmers_spp[1]), names(kmers_spp[2])),
  args.legend = list(cex = 0.7)
)
```




```
# Jaccard similarity of k-mer *presence* (not frequencies).
similarity <- length(intersect(names(a)[a > 0], names(b)[b > 0])) /
  length(union(names(a)[a > 0], names(b)[b > 0]))
similarity * 100
```

```
## [1] 80.4878
```

**Q10. Would k-mer comparisons between two or more viral isolates reflect sequence differences? Explain.**